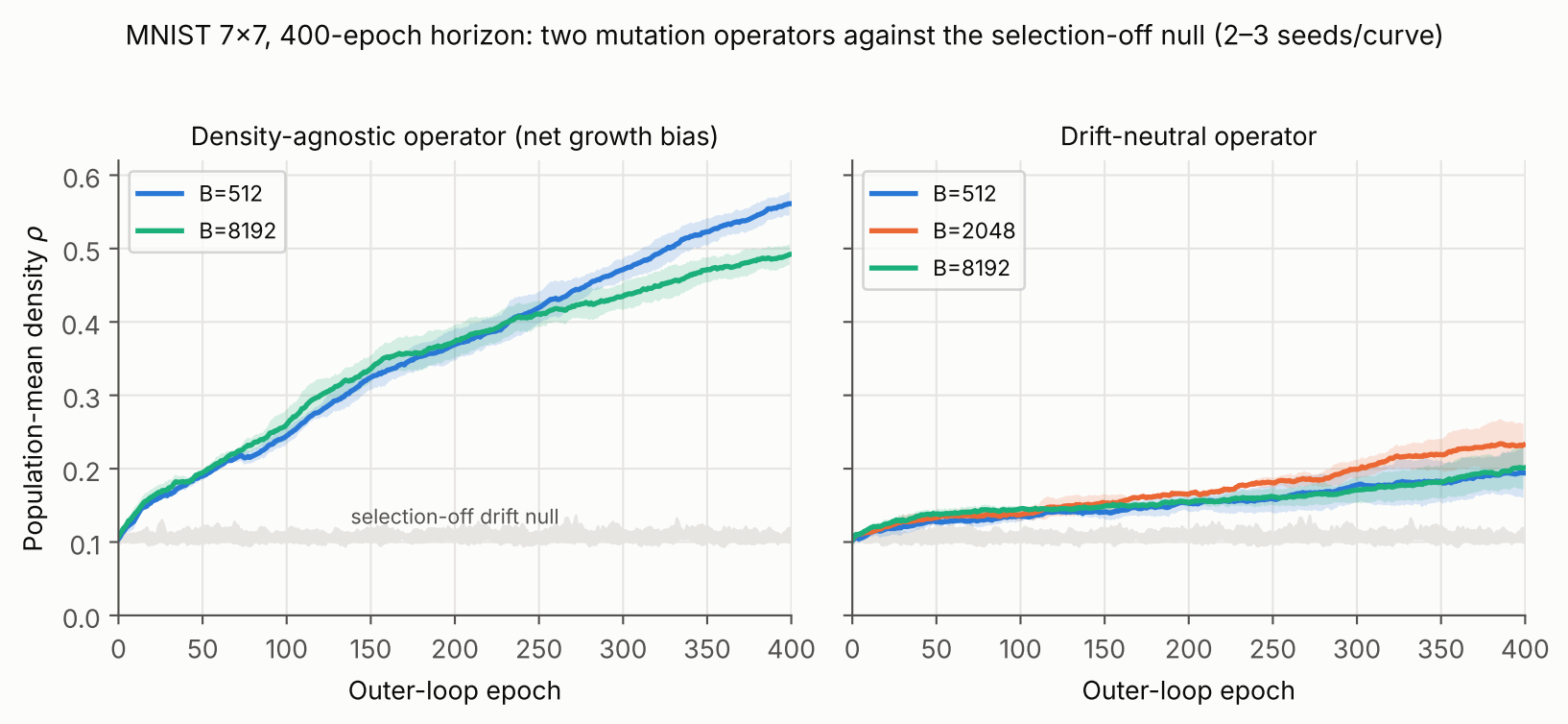

MNIST 7×7, 400-epoch horizon: two mutation operators against the selection-off null (2–3 seeds/curve)
Density-agnostic operator (net growth bias)
Drift-neutral operator
Population-mean density ρ
B=512
B=8192
B=2048
selection-off drift null
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
50
100
150
200
250
300
350
400
Outer-loop epoch